

Lattice QCD Data Summary Report

Chiral Condensates and Susceptibilities Across Finite Temperature

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Abstract

This report provides numerical measurement data tables and temperature-dependent figures for: (1) dimensionless renormalized subtracted chiral condensates $\Delta_{l,s}$, (2) temporal scaling series at $\beta = 4.17$, and (3) light-quark chiral susceptibilities χ_{scaled} [MeV²] computed with unbiased quadratic estimators across finite temperatures.

1 Definitions and Observables

1.1 Chiral Condensates

The single-flavor bare chiral condensate on a lattice volume $V = N_s^3 \times N_t$ is defined as

$$\langle \bar{\psi}\psi \rangle_q = \frac{1}{V} \langle \text{Tr} (D_q^{-1}) \rangle, \quad q \in \{l, s\} \quad (1)$$

evaluated via stochastic Z_2 volume source vectors.

The renormalized subtracted chiral condensate $\Delta_{l,s}$ eliminates additive ultraviolet divergences and residual chiral symmetry breaking:

$$\Delta_{l,s}(T) = \frac{1}{Z_m(\beta)} \left[\langle \bar{\psi}\psi \rangle_l - \frac{m_l + m_{\text{res}}}{m_s + m_{\text{res}}} \langle \bar{\psi}\psi \rangle_s \right] \quad (2)$$

where $m_{\text{res}}(\beta)$ is the residual quark mass, and $Z_m(\beta)$ is the scalar mass renormalization factor. In lattice QCD thermodynamics, $\Delta_{l,s}$ directly serves as the dimensionless order parameter for chiral symmetry restoration.

1.2 Chiral Susceptibility

The light-quark disconnected chiral susceptibility χ is defined as

$$\chi = \frac{V}{T} \left[\left\langle \left(\frac{1}{V} \text{Tr}(D_l^{-1}) \right)^2 \right\rangle - \left\langle \frac{1}{V} \text{Tr}(D_l^{-1}) \right\rangle^2 \right] \quad (3)$$

Using k stochastic volume noise vectors $O_{l,i}$ on each gauge configuration without vector-level RM subtraction, the noise bias is removed by the unbiased quadratic cross-product estimator:

$$\bar{O} = \frac{1}{k} \sum_{i=1}^k O_{l,i}, \quad \overline{O^2} = \frac{1}{k(k-1)} \sum_{i \neq j} O_{l,i} O_{l,j} = \frac{1}{k(k-1)} \left[\left(\sum_{i=1}^k O_{l,i} \right)^2 - \sum_{i=1}^k O_{l,i}^2 \right] \quad (4)$$

The ensemble susceptibility and its Jackknife statistical error are given by

$$a_r = \text{JK}(\bar{O})_r, \quad b_r = \text{JK}(\overline{O^2})_r, \quad \chi_r = b_r - a_r^2 \quad (5)$$

$$\bar{\chi} = \frac{1}{N} \sum_{r=1}^N \chi_r, \quad \sigma_\chi = \sqrt{\frac{N-1}{N} \sum_{r=1}^N (\chi_r - \bar{\chi})^2} \quad (6)$$

The continuous thermal scaled susceptibility with physical dimensions of [MeV²] is

$$\chi_{\text{scaled}} = (16T)^2 a^2 \chi = N_s^3 \times N_t^3 \times T^2 \times \chi \quad [\text{MeV}^2] \quad (7)$$

2 Chiral Condensate Data and Temperature Dependence

2.1 Numerical Data Tables

Table 1 presents the dimensionless renormalized subtracted chiral condensate $\Delta_{l,s}$ across the 8 standard temperature points on $48^3 \times 16$ lattices. Table 2 summarizes the results for the finite temporal extent scaling ensembles at $\beta = 4.17$.

Table 1: Dimensionless renormalized subtracted chiral condensate $\Delta_{l,s}$ across 8 temperature points on standard $48^3 \times 16$ lattices.

β	T [MeV]	Geometry ($N_s^3 \times N_t$)	N_{cfg}	$\Delta_{l,s}$
4.13	138.23	$48^3 \times 16$	2000	$(1.0608 \pm 0.0067) \times 10^{-3}$
4.15	145.75	$48^3 \times 16$	2024	$(7.7109 \pm 0.0675) \times 10^{-4}$
4.17	153.31	$48^3 \times 16$	2008	$(5.0728 \pm 0.0662) \times 10^{-4}$
4.18	157.03	$48^3 \times 16$	2029	$(4.1649 \pm 0.0659) \times 10^{-4}$
4.20	164.55	$48^3 \times 16$	2072	$(2.3896 \pm 0.0561) \times 10^{-4}$
4.23	176.43	$48^3 \times 16$	2012	$(9.2886 \pm 0.3810) \times 10^{-5}$
4.30	202.52	$48^3 \times 16$	2052	$(8.2043 \pm 1.3300) \times 10^{-6}$
4.405	241.60	$48^3 \times 16$	4859	$(3.4370 \pm 1.0350) \times 10^{-7}$

Table 2: Dimensionless renormalized subtracted chiral condensate $\Delta_{l,s}$ for finite temporal scaling ensembles at $\beta = 4.17$ ($m_l = 0.0020, m_s = 0.0400$).

Ensemble	Geometry	T [MeV]	N_{cfg}	$\Delta_{l,s}$
L48T18 (stream 1)	$48^3 \times 18$	136.28	98	$(8.2980 \pm 0.4422) \times 10^{-4}$
L48T18 (stream 2)	$48^3 \times 18$	136.28	76	$(7.9475 \pm 4.6593) \times 10^{-5}$
L40T16	$40^3 \times 16$	153.31	879	$(3.3260 \pm 0.7358) \times 10^{-5}$
L32T14	$32^3 \times 14$	175.21	291	$(-1.1468 \pm 1.1468) \times 10^{-8}$
L32T12	$32^3 \times 12$	204.41	108	$(-2.0497 \pm 1.8947) \times 10^{-5}$

2.2 Plots vs Temperature

2.2.1 Standard $48^3 \times 16$ Ensembles

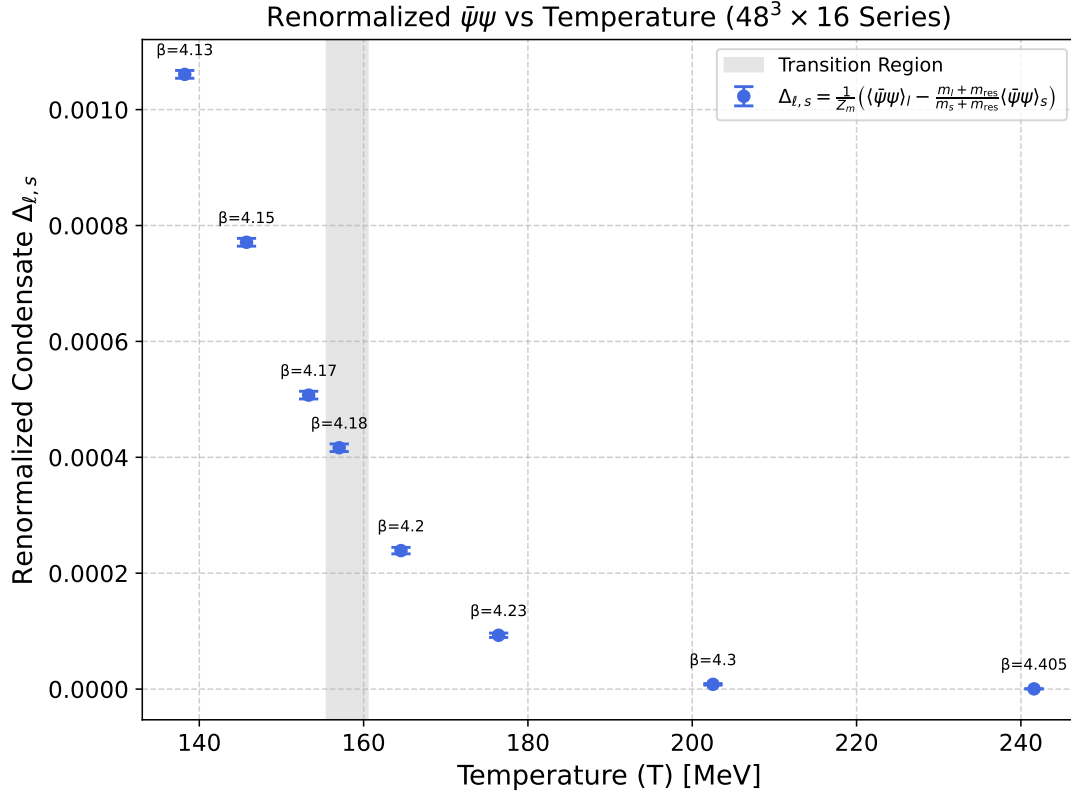


Figure 1: Dimensionless renormalized subtracted chiral condensate $\Delta_{l,s}$ as a function of temperature T [MeV] for standard $48^3 \times 16$ lattices. The shaded band marks the pseudo-critical transition window $T \in [155.5, 160.5]$ MeV.

2.2.2 Finite Temporal Scaling Ensembles ($\beta = 4.17$)

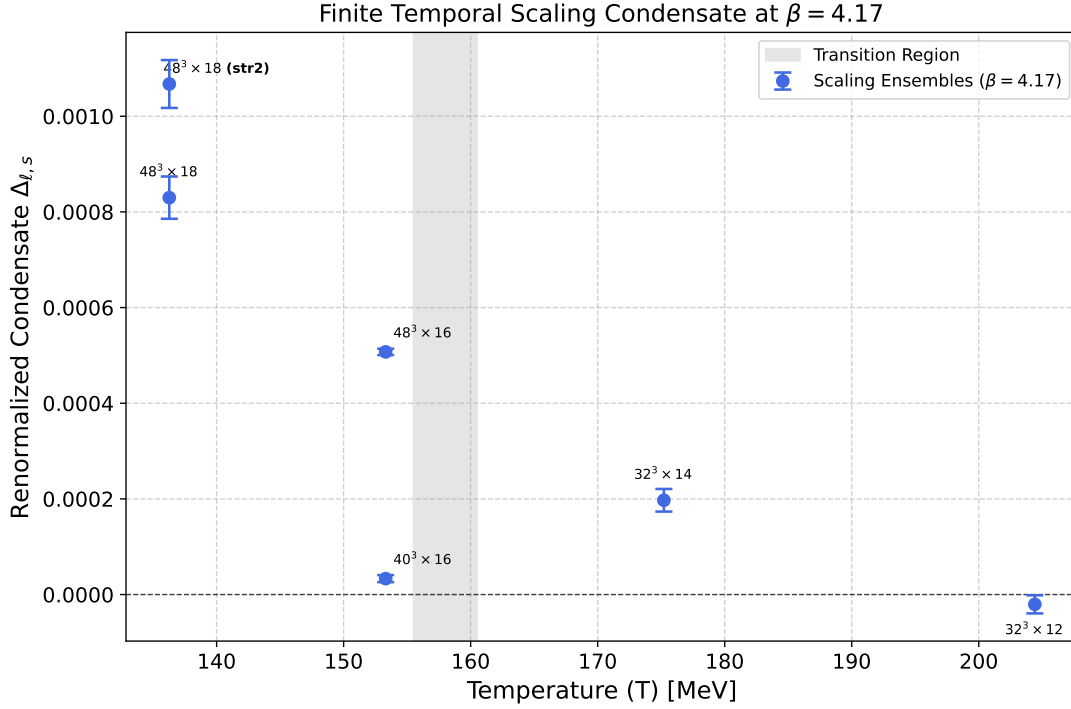


Figure 2: Dimensionless renormalized subtracted chiral condensate $\Delta_{l,s}$ across temporal geometries $N_t = 12, 14, 16, 18$ ($T = 153.31 \times 16/N_t$ [MeV]) at fixed $\beta = 4.17$.

3 Chiral Susceptibility Data and Temperature Dependence

3.1 Numerical Data Tables

Table 3 presents the continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ across the 8 standard temperature points ($48^3 \times 16$). Table 4 lists the chiral susceptibility values computed for the finite temporal scaling ensembles at $\beta = 4.17$.

Table 3: Continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ on standard $48^3 \times 16$ lattices (unbiased quadratic estimator without RM subtraction).

β	T [MeV]	Geometry ($N_s^3 \times N_t$)	N_{cfg}	$\chi_{\text{scaled}} [(16T)^2 a^2 \chi]$ [MeV ²]
4.13	138.23	$48^3 \times 16$	2000	$(6.6216 \pm 0.2307) \times 10^5$
4.15	145.75	$48^3 \times 16$	2024	$(7.8002 \pm 0.2673) \times 10^5$
4.17	153.31	$48^3 \times 16$	2008	$(8.5828 \pm 0.2577) \times 10^5$
4.18	157.03	$48^3 \times 16$	2029	$(9.1495 \pm 0.3229) \times 10^5$
4.20	164.55	$48^3 \times 16$	2072	$(7.6595 \pm 0.2769) \times 10^5$
4.23	176.43	$48^3 \times 16$	2012	$(4.0906 \pm 0.2315) \times 10^5$
4.30	202.52	$48^3 \times 16$	2052	$(7.1921 \pm 1.5027) \times 10^4$
4.405	241.60	$48^3 \times 16$	4856	$(8.6496 \pm 1.6936) \times 10^1$

Table 4: Continuous scaled chiral susceptibility χ_{scaled} for finite temporal scaling ensembles at $\beta = 4.17$ ($m_l = 0.0020, m_s = 0.0400$).

Ensemble	Geometry	T [MeV]	N_{cfg}	$\chi_{\text{scaled}} [(16T)^2 a^2 \chi]$ [MeV ²]
L48T18 (stream 1)	$48^3 \times 18$	136.28	79	$(3.1489 \pm 0.4487) \times 10^5$
L48T18 (stream 2)	$48^3 \times 18$	136.28	72	$(3.0177 \pm 0.5562) \times 10^5$
L40T16	$40^3 \times 16$	153.31	17	$(2.8070 \pm 0.8693) \times 10^5$
L32T14	$32^3 \times 14$	175.21	282	$(4.2497 \pm 0.6065) \times 10^5$
L32T12	$32^3 \times 12$	204.41	79	$(8.9135 \pm 4.9052) \times 10^4$

3.2 Plots vs Temperature

3.2.1 Standard $48^3 \times 16$ Lattices

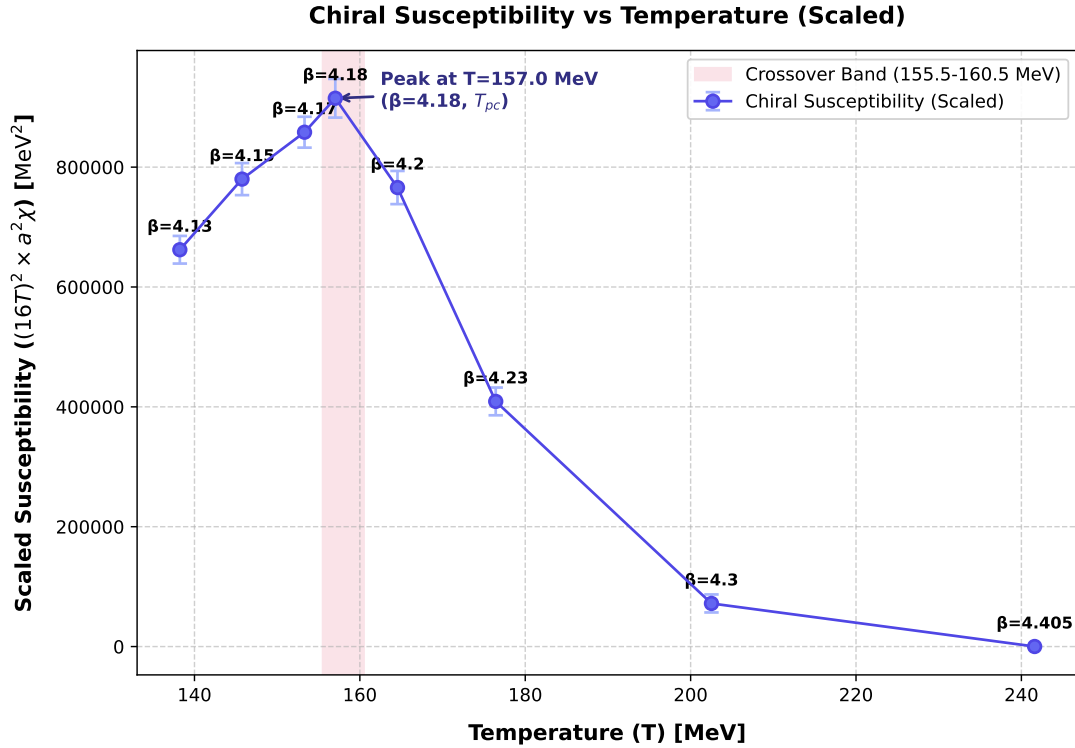


Figure 3: Continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ [MeV²] as a function of temperature T [MeV] for standard $48^3 \times 16$ lattices. The peak marks the pseudo-critical transition temperature $T_{pc} = 157.03$ MeV ($\beta = 4.18$). The shaded band denotes the crossover window $T \in [155.5, 160.5]$ MeV.

3.2.2 Finite Temporal Scaling Ensembles ($\beta = 4.17$)

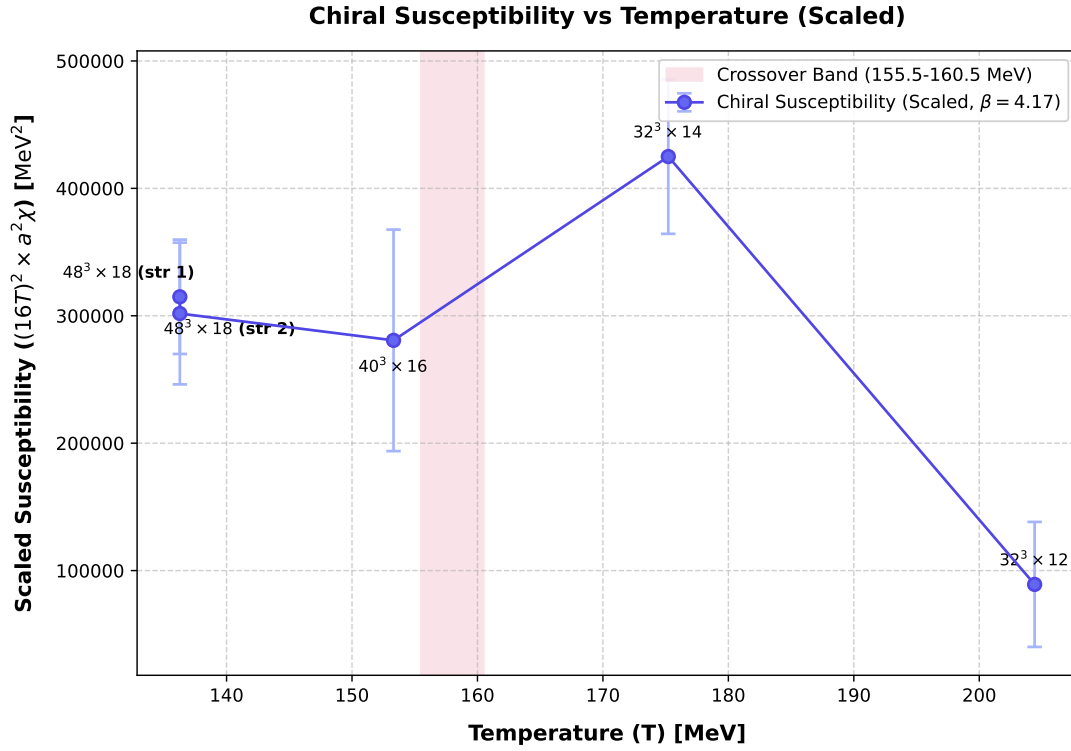


Figure 4: Continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ [MeV²] as a function of temperature T [MeV] for finite temporal scaling ensembles at $\beta = 4.17$ ($m_l = 0.0020, m_s = 0.0400$).

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